

Lesson 3 Logical Equivalence

Statements r and s are logically equivalent, denoted $r = s$, if they have the same truth table ("meaning").

Last lesson: $p \leftrightarrow q \stackrel{\text{iff}}{=} (p \rightarrow q) \wedge (q \rightarrow p)$
 $\uparrow$
 equivalence

Basic equivalences

- (Commutative Laws)

$$\rightarrow p \wedge q = q \wedge p$$

$\underbrace{\quad\quad}_V \quad \underbrace{\quad\quad}_V$

$$(p \rightarrow q \neq q \rightarrow p)$$

"not" commutative'

p	q	$p \wedge q$	$q \wedge p$
T	T	T	T
T	F	F	F
F	T	F	F
F	F	F	F

$$(a+b = b+a)$$

$ab = ba$

- (Associative laws)

$$(p \wedge q) \wedge r = p \wedge (q \wedge r)$$

$\underbrace{\quad\quad}_V \quad \underbrace{\quad\quad}_V \quad \underbrace{\quad\quad}_V \quad \underbrace{\quad\quad}_V$

aside: $+ \leftrightarrow V$
 $\times \leftrightarrow \wedge$

$$(a+b)+c = a+(b+c)$$

$\begin{matrix} x & x & & x & x \end{matrix}$

Duality betw $\wedge$ and $\vee$

- (Distributive laws)

$$p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$$

$\underbrace{\quad\quad}_V \quad \underbrace{\quad\quad}_\wedge \quad \underbrace{\quad\quad}_V \quad \underbrace{\quad\quad}_\wedge \quad \underbrace{\quad\quad}_V \quad \underbrace{\quad\quad}_V$

$$a(b+c) = ab+ac$$

$$a+(bc) = (a+b) \cdot (a+c)$$

not $\times$

- (Idempotent)

$$\begin{aligned} \rightarrow & \quad p \wedge p = p \\ \rightarrow & \quad p \vee p = p \end{aligned} \quad \}$$

idempotent is an element which is unchanged when operated by itself

mult : $0 \times 0 = 0$

$$1 \times 1 = 1$$

2 idempotents : 0 and 1

$a \times a \neq a$ in general

$$\begin{aligned} p \wedge T &= p \\ p \vee F &= p \end{aligned} \quad \} \text{ identity laws}$$

$$\begin{aligned} p \wedge F &= F \\ p \vee T &= T \end{aligned} \quad \} \text{ domination}$$

→ $p \wedge \sim p = F$
 contradiction, always false

→ $p \vee \sim p = T$
 tautology, always true
 thms are true, but not
 tautologies

• Double negatives

$$\sim(\sim p) = p$$

p	$\sim p$	$\sim(\sim p)$
T	F	T
F	T	F

• De Morgan's Laws

$$\sim(p \wedge q) = \sim p \vee \sim q$$

p	q	$p \wedge q$	$\sim(p \wedge q)$	$\sim p$	$\sim q$	$\sim p \vee \sim q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

x is positive and x is ~~negative~~ ^{even}
 negation: x is not positive or x is not ~~negative~~ ^{even}
 $\neg(p \wedge q) \neq \neg p \wedge \neg q$

- ① use T.T.
- ② use a example/counterexample

$$p = T, q = F$$

$$\neg(p \wedge q) = \neg(T \wedge F) = \neg F = T \neq$$

$$\neg p \wedge \neg q = \neg T \wedge \neg F = F \wedge T = F$$

• Contrapositive ←

$$p \rightarrow q = \neg q \rightarrow \neg p$$

T.T.
exercise

$$(p \rightarrow q \neq q \rightarrow p, p \rightarrow q \neq \neg p \rightarrow \neg q)$$

if $\underbrace{x \text{ is not even}}_p$, then it is $\underbrace{\text{odd}}_q$

contrapositive: if x is not odd, then x is even

• Conditional as "or"

$$p \rightarrow q = \sim p \vee q$$

proof: T.T., notes

→ same as: $\underbrace{x \text{ is even}}_{\sim p} \text{ or } \underbrace{x \text{ is odd}}_q$

• Negation of conditional ↙

$$\begin{aligned} \sim(p \rightarrow q) &= \sim(\sim p \vee q) \\ &\stackrel{\text{De Morgan}}{=} \sim(\sim p) \wedge \sim q \\ &\stackrel{\text{(double negatives)}}{=} p \wedge \sim q \end{aligned}$$

$(p \rightarrow \sim q) \times$

① use TT to prove

② we used previously established equivalences to prove

If $\underbrace{x \text{ is not even}}_P$, then $\underbrace{x \text{ is odd}}_Q$.

negation: $x \text{ is not even}$ and $\neg Q$ $x \text{ is not odd.}$

- ("and" implies "or")

$$(P \wedge Q) \rightarrow P$$

$$(P \wedge Q) \rightarrow Q$$

$$(P \wedge Q) \rightarrow P \vee Q$$

$$P \rightarrow P \vee Q \quad \checkmark$$

$$P \not\rightarrow P \wedge Q$$

tautology

prove
using
t.t.
